Mittuniversitetet
MID SWEDEN UNIVERSITY

Försättsblad Prov Original

Kurskod	Provkod	Tentamensdatum
MA080G	T102	2024-08-27
Kursnamn	Matematik GR (B), Kryptografi	
Provnamn	Skriftlig tentamen	
Ort	Sundsvall	
Termin		
Ämne		

6895642268

Utskriven 2024-09-23 kl. 13:04:06

Tentamen
MA080G Kryptografi
27 August 2024

Mittuniversitetet
IMD
Aurélio Menegon

Skrivtid: 5 timmar

Hjälpmedel: Miniräknare som ej är symbolhanterande. Ange märke och modell på din miniräknare på omslaget till tentamen. Inga böcker, anteckningar eller mobiltelefoner är tillåtna i tentamensrummet. This is a closed book examination. No books, notes or mobile telephones are allowed in the examination room. Electronic calculators may be used provided they cannot handle formulas. The make and model used must be specifed on the cover of your script.

Till alla uppgifter skall fullständiga lösningar lämnas. Resonemang och uträkningar får inte vara så knapphändiga att de blir svåra att följa. En uppgift per blad, skriv endast på en sida.

Betyg sätts efter hur väl lärandemålen är uppfyllda. Riktvärden för betygen är A 22p, B 18p, C 14p, D 10p och E 9p.

---

1. Answer the following questions. **(4p)**
   (a) The message
   `FUBS WRJUD SKBLV IXQ.`
   is a ciphertext encrypted using a Ceaser cipher. What is the key?
   (b) Eve gets a message, and she knows it is a ciphertext encrypted using a Vigenère cipher. Using the Friedman's attack, she estimates that the length of the key is 4. She also has the letter frequencies by position modulo 4 (in the table on the next page). What is the key? Justify your answer.
2. Find all the generators for the cyclic group $(\mathbb{Z}/8\mathbb{Z}, +)$. **(2p)**
3. If $n = 539$, how many units does $\mathbb{Z}/n\mathbb{Z}$ have? **(2p)**
   Hint: Recall that $\phi(p^e) = p^e - p^{e-1}$, where ϕ denotes Euler's totient function.
4. Use Euler's generalization of Fermat's little theorem to calculate $3^4 \pmod 8$. **(2p)**
5. Recall the lemma that supports the AKS Primality test:
   If $gcd(a, n) = 1$, then n is prime if and only if $(x+a)^n \equiv x^n + a \pmod n$.
   Using the lemma, verify that 6 is not prime. **(2p)**
6. Show that $g = 5$ is a primitive root of $\mathbb{F}_p^\times$ when $p = 37$. **(2p)**
7. Your private key is $a = 12$. Use the *Diffie-Hellman Key exchange method* to create your public key, with $p = 19$ and $g = 2$. Use the fast exponentiation algorithm on your computation. **(2p)**
8. The cryptosystem is RSA. Bob's public key is $n = 2773$ and $e = 3$. What is Bob's private key? **(4p)**

9. Alice and Bob are such good friends that they choose to use RSA with the same n, but their encryption exponents e and f are different, and indeed, they are relatively prime. Charles wants to send the same message m to Alice and Bob. If Eve intercepts both of his messages, how can she recover the plaintext message m? **(4p)**

Table of letter frequency analysis for Question 1:

Letter	**1**	**2**	**3**	**4**
A	5	3	4	0
B	5	2	2	1
C	6	6	1	0
D	6	2	18	0
E	2	4	4	17
F	1	5	3	1
G	1	3	4	1
H	0	5	5	4
I	0	3	2	16
J	0	2	1	2
K	0	1	5	1
L	16	0	4	1
M	4	1	2	18
N	0	1	0	2
O	5	16	2	4
P	14	1	2	3
Q	1	2	0	3
R	3	0	1	5
S	2	14	1	4
T	15	1	0	1
U	5	5	0	5
V	0	3	16	5
W	2	13	4	1
X	4	1	2	2
Y	1	2	2	1
Z	2	4	12	0